5. Calculate correlation coefficient for the following data

X	25	18	21	24	27	30	36	39	42	48
Y	26	35	48	28	20	36	25	40	43	39

6. Find coefficient of correlation for the following:

X	78	89	96	69	59	79	68	62
Y	121	72	88	60	81	87	123	92

7. An examination of 11 applicants for a accountant post was taken by a finance company. The marks obtained by the applicants in the reasoning and aptitude tests are given below.

Applicant	A	B	C	D	E	F
Reasoning test	20	50	28	25	70	90
Aptitude test	30	60	50	40	85	90

Applicant	G	H	I	J	K
Reasoning test	76	45	30	19	26
Aptitude test	56	82	42	31	49

Calculate Spearman's rank correlation coefficient from the data given above.

8. The following are the ranks obtained by 10 students in commerce and accountancy are given below:

Commerce	6	4	3	1	2	7	9	8	10	5
Accountancy	4	1	6	7	5	8	10	9	3	2

To what extent is the knowledge of students in the two subjects related?

9. A random sample of recent repair jobs was selected and estimated cost and actual cost were recorded.

Estimated cost	300	450	800	250	500	975	475	400
Actual cost	273	486	734	297	631	872	396	457

Calculate the value of spearman's correlation coefficient.

10. The rank of 10 students of same batch in two subjects *A* and *B* are given below. Calculate the rank correlation coefficient.

Rank of A	1	2	3	4	5	6	7	8	9	10
Rank of B	6	7	5	10	3	9	4	1	8	2

9.3 Regression Analysis

Introduction:

So far we have studied correlation analysis which measures the direction and strength of the relationship between two variables. Here we can estimate or predict the value of one variable from the given value of the other variable. For instance, price and supply are correlated. We can find out the expected amount of supply for a given price or the required price level for attaining the given amount of supply.

The term " regression" literally means "Stepping back towards the average". It was first used by British biometrician Sir Francis Galton (1822 -1911), in connection with the inheritance of stature. Galton found that the offsprings of abnormally tall or short parents tend to "regress" or "step back" to the average population height. But the term "regression" as now used in Statistics is only a convenient term without having any reference to biometry.

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